

Product Portfolio & Margin Intelligence Dashboard

Built by Josh Tacker | 2026

Tech Stack: Python · PostgreSQL · R Shiny · Plotly

Data: Global Superstore 2011–2014 | 10,768 SKUs | \$12.6M Revenue

The Problem Every Pricing Manager Faces

28%

of SKUs

are losing money
(negative margin)

1.1%

of Revenue

from tail-end SKUs
(portfolio complexity)

7.7%

Avg Margin

with huge variation
across sub-categories

Without the right tools, pricing managers are flying blind.

This dashboard gives them the visibility to act.

Dashboard Architecture — 5 Analytical Views

01

Portfolio Overview

KPIs, revenue mix,
monthly trend

02

Margin Management

Alerts, margin by
sub-category

03

Price Optimization

Price vs margin
scatter + discounts

04

Channel & Region

Segment, region,
ship mode margin

05

SKU Explorer

Filter & drill into
individual SKUs

Global filters (Category, Segment, Year) apply across all tabs simultaneously

Tab 1: Portfolio Overview

\$12.6M

Total Revenue

\$1.5M

Total Profit

11.6%

Avg Margin

3,027

Neg. Margin SKUs

2,154

Tail-End SKUs

10,768

Total SKUs

What the Manager Sees

- Revenue & profit by sub-category side by side
- Revenue mix pie chart by category
- Monthly trend line — spot seasonality instantly
- All 6 KPIs update when filters change

Why It Matters for AkzoNobel

- Product managers need a daily health check at a glance
- Seasonal patterns inform promotional planning
- The KPI cards surface problems before they escalate
- Identical workflow to PMR cycle reporting

Tab 2: Margin Management

Negative Margin Alert

3,027 SKUs (28% of portfolio) are losing money. They represent \$2.1M in revenue at negative margin.

Tail-End Rationalization

2,154 tail-end SKUs generate only 1.1% of total revenue. Significant complexity cost.

Best Opportunity

Shift mix toward Paper (24.2%) and away from Tables (-8.5%). Estimated margin lift: +3-5pp.

Margin % by Sub-Category

Horizontal bar chart — red/orange/green color coding. Tables at -8.5%, Paper at +24.2%. Immediate visual priority.

Discount Impact on Margin

As discount band increases, margin collapses to negative. Visual proof that deep discounting destroys value.

Margin Distribution

Histogram of all 10,768 SKUs by margin. Break-even line at 0%, avg line at 7.7%. Shows the full picture of portfolio health.

Tab 3: Price Optimization

Price vs. Margin Scatter Plot

Every dot is a SKU. Bubble size = Revenue. Color = Sub-Category.

- HIGH PRICE / HIGH MARGIN — Stars — protect and grow
- LOW PRICE / HIGH MARGIN — Price increase opportunity
- HIGH PRICE / LOW MARGIN — Cost structure problem
- LOW PRICE / LOW MARGIN — Rationalization candidate

Discount Band Analysis

Dual-axis chart: revenue bar + margin line.
Margins go negative at >30% discount.
Quantifies the exact cost of each discount level.

Sub-Category Opportunity Matrix

Bubble chart: discount rate vs margin.
Large bubbles = high revenue. Immediately identifies where to cut discounts first.

Tabs 4 & 5: Channel Intelligence + SKU Explorer

Tab 4: Channel & Region

► Margin by Customer Segment

Consumer vs Corporate vs Home Office — which channel to prioritize for pricing strategy

► Revenue vs Margin by Region

Bubble chart — high-revenue regions with low margin are your biggest pricing opportunity

► Ship Mode Margin Analysis

Which fulfillment mode is most profitable — informs pricing architecture by channel

► Top 15 Countries by Revenue

With margin % shown — identifies global markets for price optimization focus

Tab 5: SKU Explorer

Four filter modes:

All SKUs

Full portfolio view with sortable columns

Negative Margin

Instantly surfaces the 3,027 loss-making SKUs

Tail-End Only

Shows the 2,154 candidates for rationalization

High Margin >20%

Identifies star products to protect and grow

Color-coded margin column: red < 0% | orange 0-10% | green > 10%

How It Was Built



Key decisions:

R Shiny chosen for reactive filtering · Plotly for interactive hover details · Color coding maps to traffic light pricing standards

Key Findings — Actionable Insights

01

Tables is destroying margin

At -8.5% average margin, Tables sub-category is the single largest margin drag. Every unit sold loses money. Immediate price increase or cost review required.

02

Discounts above 30% always result in negative margin

Regardless of product category, any discount > 30% produces negative margin. This gives pricing teams a hard guardrail: no discount above 30% without VP approval.

03

Paper is the star — protect and grow

Paper delivers 24.2% margin at high volume. It is underweighted in the portfolio. Mix shift toward Paper and away from low-margin sub-categories improves overall margin.

04

2,154 tail-end SKUs generate only 1.1% of revenue

Eliminating or consolidating tail-end products reduces complexity cost, frees up working capital, and focuses pricing attention on what actually moves the needle.

Why This Matters for AkzoNobel

Dashboard Capability

Margin % by sub-category with alert thresholds

Tail-end SKU identification and rationalization

Discount band impact on margin

Price vs margin scatter — find mispriced products

Regional margin variance

Mix shift recommendations

AkzoNobel Application

→ Wood Finishes vs Sikkens vs Dulux margin monitoring

→ Portfolio complexity reduction — fewer SKUs, higher margins

→ PMR cycle guardrails — price floors by product line

→ Win room analysis — price architecture by segment

→ Geographic pricing strategy — NA vs EMEA vs APAC

→ Customer mix optimization for improved portfolio margin

What I'd Build Next

- ▶ Connect to live AkzoNobel data sources — SAP, ERP, or data warehouse
- ▶ Add price elasticity modeling — predict revenue impact of price changes
- ▶ Build a PMR cycle tracker — monitor pricing decisions and outcomes over time
- ▶ Competitor price benchmarking — add external market data for positioning
- ▶ Automated margin alert emails — notify product managers when SKUs cross thresholds

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